NBS4MED self-evaluation report

Generated: 2026-10-17

Demo version: illustrative estimates, to be reviewed when project data become available

Preliminary prototype results for the area as it is today. The methodology will be reviewed against the PROMEA framework and the NBS repository. The scores are NBS4MED prototype scores and are not an official score or certification of any external organisation or standard.

1. Assessment area

Assessment area	İzmir Green Roof Pilot
Country	Türkiye
City / municipality	İzmir
Assessment scale	Site / Project Area
Organisation	Not stated
Assessment date	2026-10-17

2. Assessment of the five criteria (provisional)

The assessment describes existing NBS integration (C1) and the enabling and implementation conditions (C2 to C5). It is provisional and separate from the suitability, impact and cost of a solution.

Completeness

9 of 14 indicators have a numeric score. 2 Unknown. 3 Not applicable.

Criterion profile

Current NBS integration (C1). How far NBS are already integrated in the area today. This criterion does not measure environmental effectiveness, such as cooling or flood reduction.

Criterion	Score	Level	Completeness and confidence
C1 NBS Integration Status	Not rated: Unknown answers	–	2 of 3 indicators scored · 1 Unknown. Available points only (not a criterion score): 1 of 4 on 2 of 3 indicators

Enabling and implementation conditions (C2 to C5). Policy, capacity, engagement and funding that support planning, delivery and long-term upkeep of NBS.

Criterion	Score	Level	Completeness and confidence
C2 Policy & Planning	2 / 4 (50%)	Partial	2 of 3 indicators scored · 1 Not applicable
C3 Institutional & Technical Capacity	Not rated: Unknown answers	–	2 of 3 indicators scored · 1 Unknown. Available points only (not a criterion score): 3 of 4 on 2 of 3 indicators
C4 Stakeholder Engagement	Not applicable	Not applicable	0 of 2 indicators scored · 2 Not applicable
C5 Funding & Sustainability	3 / 6 (50%)	Partial	3 of 3 indicators scored

Provisional combined assessment score (secondary; not a validated measure of NBS integration)

The overall percentage is not shown because 2 indicators are Unknown (1.2, 3.3). Resolve them to see the overall score.

The overall score gives every indicator equal weight, so a criterion with more indicators counts for more. Scored indicators per criterion: C1 2, C2 2, C3 2, C4 0, C5 3. Provisional project thresholds, not taken from any external standard: Insufficient below 30%, Partial 30 to 54%, Adequate 55 to 79%, Strong 80% and above. Project partners will review them.

Gaps and strengths by indicator

Priority gaps (score 0)	1.3 Link to grey infrastructure: NBS are planned separately from drainage, roads and buildings, or not linked to them at all. 5.3 Co-benefits valued in decisions: Co-benefits such as cooling, health and flood protection are not counted in funding decisions.
Developing (score 1)	1.1 NBS inventory: An inventory exists but is incomplete or needs updating. 2.2 Regulation and standards: Some rules or standards encourage NBS, but they are voluntary, partial or rarely applied. 2.3 Targets and monitoring: Targets exist but are not measurable, or progress is not reported regularly. 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support. 5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS.
Strengths (score 2)	3.1 Dedicated staff or unit 5.2 Long-term maintenance budget
Information gaps (Unknown)	1.2 Coverage of public space 3.3 Data and monitoring tools
Not applicable	2.1 NBS in strategic plans: Not relevant to this site: no residents live in the project area 4.1 Community participation: Not relevant to this site: no residents live in the project area 4.2 Coordination across departments and owners: Not relevant to this site: no residents live in the project area

Priority gaps and improvement actions

Shown in a provisional display order: answers scored 0 first, then 1, then Unknown, each in indicator order. This is not a validated ranking of impact or urgency. The actions are general prompts based on your answers, not findings about the site.

#	Indicator	Type	Finding	Why it matters	Suggested action	Evidence needed
1	1.3 — Link to grey infrastructure	Gap	NBS are planned separately from drainage, roads and buildings, or not linked to them at all.	NBS planned with drainage, roads and buildings are cheaper to build, easier to maintain and more likely to work as intended.	Introduce a step in infrastructure projects to consider NBS at the design stage.	Project design documents, drainage or road plans and procedures for infrastructure projects.
2	5.3 — Co-benefits valued in decisions	Gap	Co-benefits such as cooling, health and flood protection are not counted in funding decisions.	Counting co-benefits can strengthen the case for funding NBS compared with grey alternatives.	Include a short co-benefit description in funding proposals and business cases.	Funding proposals, business cases and any co-benefit studies.
3	1.1 — NBS inventory	Partial	An inventory exists but is incomplete or needs updating.	Without a record of what exists, the area cannot plan, maintain or build on its NBS.	Update and complete the inventory so that it covers the whole area and records type, location and condition.	GIS inventories, municipal records or site documentation.
4	2.2 — Regulation and standards	Partial	Some rules or standards encourage NBS, but they are voluntary, partial or rarely applied.	Rules and incentives make NBS a normal part of development, not a voluntary extra.	Clarify and apply the existing rules or incentives more consistently, and track how often they are applied.	Zoning regulations, building codes, green-area standards and examples of their application.

#	Indicator	Type	Finding	Why it matters	Suggested action	Evidence needed
5	2.3 — Targets and monitoring	Partial	Targets exist but are not measurable, or progress is not reported regularly.	Measurable targets and regular reporting show whether NBS are progressing and allow plans to be corrected.	Make the existing targets measurable and set a regular schedule for reporting progress.	Target statements, monitoring indicators and progress reports.
6	3.2 — Technical know-how	Partial	Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.	Technical know-how decides whether NBS are built well and kept working over time.	Extend skills and guidance to further NBS types and reduce reliance on outside support.	Examples of delivered projects, guidance documents and staff or contractor qualifications.
7	5.1 — Capital funding	Partial	Some funding is secured or likely, but it does not cover the planned NBS.	Without capital funding, planned NBS stay on paper.	Close the funding gap for the planned NBS by securing additional sources or phasing the works.	Budget documents, grant agreements and funding commitments.
8	1.2 — Coverage of public space	To verify	No information is available for this indicator yet. This is a gap in information, not evidence of weak performance.	Coverage shows how much of the public space can already contribute to cooling, shade and water management.	Obtain or verify this information before drawing conclusions.	Land-cover or green-space maps, municipal green-area statistics and any local target.
9	3.3 — Data and monitoring tools	To verify	No information is available for this indicator yet. This is a gap in information, not evidence of weak performance.	Data are needed to place NBS where they help most and to check whether they work.	Obtain or verify this information before drawing conclusions.	Dataset lists, data-sharing agreements and examples of data used in decisions.

Confidence in the answers

Of the 9 scored indicators: 1 documented evidence, 0 staff estimate, 0 informal judgement, 8 confidence not stated. Confidence is reported next to the answers. It does not change any score.

Answers in detail

ID	Indicator	Answer	Rationale	Evidence or source	Confidence
1.1	NBS inventory	1 of 2	Based on the municipal GIS layer	Inventory report 2024	High: documented evidence
1.2	Coverage of public space	Unknown	–	–	Not stated
1.3	Link to grey infrastructure	0 of 2	–	–	Not stated
2.1	NBS in strategic plans	Not applicable	Not relevant to this site: no residents live in the project area	–	Not stated
2.2	Regulation and standards	1 of 2	–	–	Not stated
2.3	Targets and monitoring	1 of 2	–	–	Not stated
3.1	Dedicated staff or unit	2 of 2	–	–	Not stated
3.2	Technical know-how	1 of 2	–	–	Not stated
3.3	Data and monitoring tools	Unknown	–	–	Not stated
4.1	Community participation	Not applicable	Not relevant to this site: no residents live in the project area	–	Not stated
4.2	Coordination across departments and owners	Not applicable	Not relevant to this site: no residents live in the project area	–	Not stated
5.1	Capital funding	1 of 2	–	–	Not stated
5.2	Long-term maintenance budget	2 of 2	–	–	Not stated

ID	Indicator	Answer	Rationale	Evidence or source	Confidence
5.3	Co-benefits valued in decisions	0 of 2	–	–	Not stated

Is there a local target for NBS or for green and blue space coverage? No. Is the current coverage known (mapped or measured)? No.

Site context (not scored)

Urban Density	High
Public Land Available for NBS	Limited
Local Climate Risks	Heatwaves; Flooding

How the assessment scores are calculated (provisional)

- Each indicator is scored 0, 1 or 2 using the provisional project rubric shown in the assessment.
- Unknown is missing information and is never counted as 0. Not applicable (with a justification) is also left out. Both are reported separately.
- A criterion percentage is the sum of its scored indicators divided by 2 times the number of scored indicators. It is shown only when the criterion has no Unknown answer. A criterion whose indicators are all Not applicable is shown as Not applicable.
- The overall percentage uses the same rule across all indicators and is shown only when no indicator is Unknown. With every indicator scored, it is the total divided by 28. Every indicator has equal weight, so criteria with more indicators weigh more.
- Levels use provisional project thresholds: Insufficient below 30%, Partial 30 to 54%, Adequate 55 to 79%, Strong 80% and above.
- The scores are not multiplied by confidence. Confidence is reported separately, indicator by indicator.
- C1 describes current NBS integration only. It is not a measure of environmental effectiveness.

3. Applicability screening

Screening only, not engineering approval. These checks do not change the assessment score or the match score. Answers about planting space, soil and water are shared by the planting-based options.

No solution has been checked yet. Applicability statuses show Not yet checked.

4. Candidate solutions and reasons

Options are listed in the order of a provisional rule-based match to your selected risks and site context. This is not a validated ranking. Applicability statuses come from optional checks and are not engineering approval. The match is a provisional rule-based description of how well an option addresses the selected risks and the land context. It is not a probability of success, a feasibility assessment or a predicted climate impact.

#	Option	Category (provisional)	Rule-based match score	Screening status	Why it matches the selected risks	To verify	Implementation conditions
1	Green roof	Green infrastructure	67 / 100	Not yet checked	Heatwaves (moderately); Flooding (moderately). Uses existing roof space, not public land. Roof availability and structure still need checking.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
2	Green corridor	Green infrastructure	67 / 100	Not yet checked	Heatwaves (strongly); Flooding (weakly). Land need fits the available public land.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
3	Urban forest (Miyawaki method)	Green infrastructure · Ecosystem restoration	52 / 100	Not yet checked	Heatwaves (strongly); Flooding (weakly). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
4	Rain garden and bioswale	Blue-green water management	52 / 100	Not yet checked	Heatwaves (weakly); Flooding (strongly). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
5	Permeable paving	Blue-green water management · Hybrid/integrated solutions	50 / 100	Not yet checked	Flooding (strongly). Not a target of this option: Heatwaves. Land need fits the available public land.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.

#	Option	Category (provisional)	Rule-based match score	Screening status	Why it matches the selected risks	To verify	Implementation conditions
6	Smart eco-park	Hybrid/integrated solutions · Green infrastructure	43 / 100	Not yet checked	Heatwaves (strongly); Flooding (moderately). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
7	Pocket park	Green infrastructure	35 / 100	Not yet checked	Heatwaves (moderately); Flooding (weakly). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
8	Rainwater retention basin	Blue-green water management · Hybrid/integrated solutions	27 / 100	Not yet checked	Heatwaves (weakly); Flooding (strongly). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
9	Vegetated dune and shoreline	Ecosystem restoration	2 / 100	Not yet checked	Flooding (weakly). Not a target of this option: Heatwaves. Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.

How the rule-based match score is calculated (provisional)

- Each option has a weight from 0 (not a target) to 3 (strong) for heatwaves, flooding and drought. These are the earlier demo weights, kept as provisional rules pending methodology review. They have not been validated by PROMEA.
- Matching rules for Coastal Storms and Land Degradation have not been defined. When either is selected, or when Not Sure is selected, no match score or ranking is produced.
- Score = sum of the weights for your selected risks, divided by 3 times the number of selected risks, times 100.
- Land context: 15 points are subtracted for each level by which the option's land need exceeds the public land available, and 10 more points when urban density is high and the land need is high. The score is limited to 0 to 100.

- Urban density and public land availability are qualitative answers, not numerical thresholds. If public land availability is Unknown, no match score is produced. Institutional gaps (funding, maintenance, technical know-how) are not used to change the score. They are shown as implementation conditions.
- Climate type is not yet used in matching. It will be considered once verified location-specific information and decision rules are available.
- Public land availability is not used as evidence that a roof is available. Roofs are checked separately in the applicability checks.
- Screening: an option is Not applicable only when a basic condition is answered No (for example no roof, no coast, no runoff source). Unknown, or a No on a condition that can be assessed or designed (for example structural suitability), gives Requires verification.

Option	Heatwaves	Flooding	Drought
Urban forest (Miyawaki method)	3	1	1
Green roof	2	2	1
Green corridor	3	1	0
Rain garden and bioswale	1	3	1
Permeable paving	0	3	1
Pocket park	2	1	0
Smart eco-park	3	2	2
Rainwater retention basin	1	3	3
Vegetated dune and shoreline	0	1	0

Working alignment with the ten PROMEA considerations

Working framework used by the project team. This is not a formal PROMEA validation. Items marked Pending are not yet operational in this prototype.

Consideration	Status	Detail
Climate Risks	Partially supported	Selection of five risks (multiple allowed, plus Not Sure) is implemented. Provisional matching rules exist only for heatwaves, flooding and drought.
Climate Type	Pending further development	Not asked and not inferred. Needs verified location-specific information and decision rules.
Urban Context	Partially supported	Urban density and public land availability are asked as qualitative answers and used in the provisional matching.
NBS Categories	Partially supported	Provisional category labels on each option. They do not affect matching.
Cost	Partially supported	Indicative demo cost ranges for five of the nine options, not local prices.
Stakeholders	Partially supported	C4 Stakeholder Engagement (4.1, 4.2) on the provisional 0 to 2 rubric. Not used in matching.
Regulations	Partially supported	C2 Policy & Planning (2.1 to 2.3) on the provisional 0 to 2 rubric. Not used in matching.
Technical Expertise	Partially supported	C3 Institutional & Technical Capacity (3.1 to 3.3). Technical know-how (3.2) is also shown as an implementation condition.
Maintenance	Partially supported	Indicator 5.2, shown as an implementation condition, and an annual maintenance cost range for supported options.
Evaluation	Pending further development	Indicators 2.3 and 3.3 are assessed, but there is no monitoring or evaluation module for NBS performance.

5. Impact and cost exploration (illustrative)

The impact and cost step was not completed. Only the assessment is reported.

Demo version: illustrative estimates, to be reviewed when project data become available